

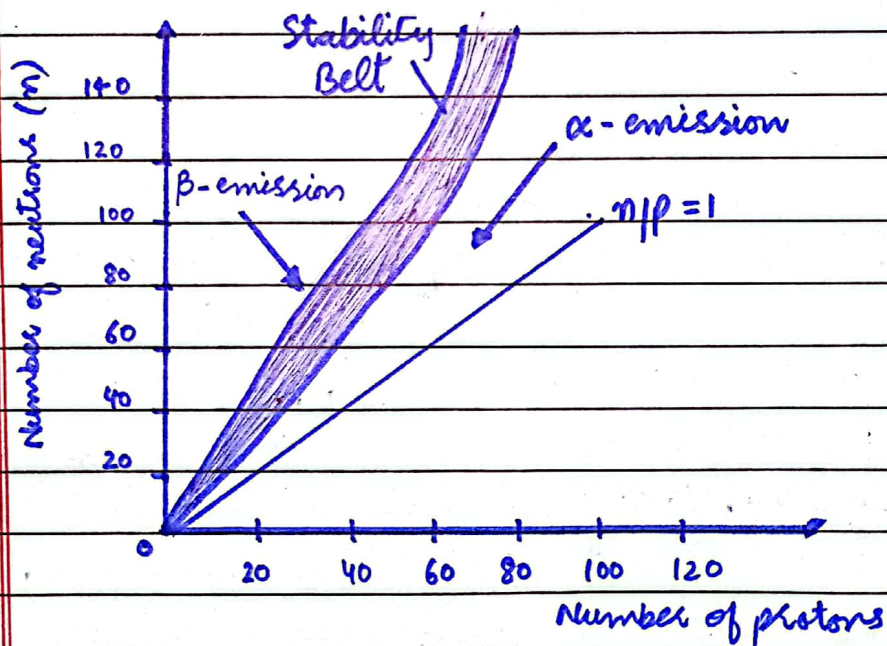
2023

Short Answers

(1)

How neutron to proton ratio is related to the stability of nucleus.

The stability of nucleus depends on the number of protons and neutrons present. The following graph shows the neutron-to-proton ratios for all known stable elements.



Each point on the graph indicates the number of protons and neutrons in a particular stable nucleus. It is clear from the graph that:

- (1) The lower elements (upto $Z=30$), having the stable nuclei, have about

equal number of protons and neutrons
i.e., $n/p = 1$.

(2) For higher elements to be stable, there must be more neutrons than protons
i.e., $n/p > 1$

(3) The shaded portion in graph represents the region or belt of stability. The elements whose n/p lie inside the belt are stable.

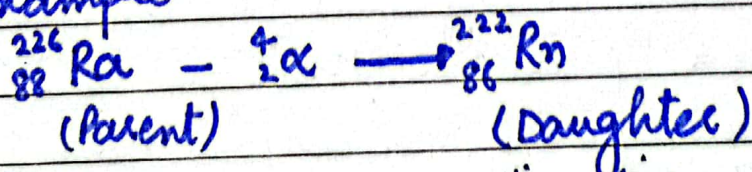
(4) A nucleus whose n/p lies above or below the stability belt is radioactive or unstable on account of unfavourable n/p ratio. It emits α or β particles so as to move into stability range.

(2)

Different types of radioactive decay.

There are two chief types of radioactive decay

1. **α -Decay:** When a radioactive nucleus decays by the emission of an α -particle from the nucleus, the process is termed as α -decay.
For example

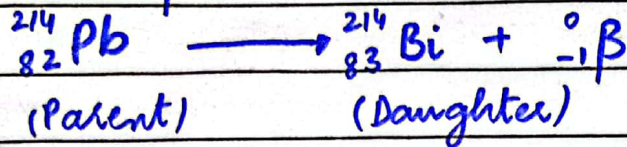


2. **β -Decay:** When a radioactive nucleus decays by β -particle emission, it is

called β -decay.

A β emission increases the atomic number by 1 with no change in atomic mass.

For example



(3)

Nuclear Fission

- A bigger (heavier) nuclei splits into smaller (lighter) nuclei.

- It does not require high temperature.

- A chain reaction sets on.

- The energy liberated during the process is high (200 MeV per fission).

- It can be controlled and energy released can be used for peaceful purposes.

Nuclear Fusion

- Lighter nuclei fuse together to form the heavier nucleus.

- Extremely high temperature (10^8K) is required for fusion to take place.

- It is not a chain reaction.

- The energy liberated during the process is comparatively low (3-24 MeV per fusion).

- It can not be controlled and energy released can not be used properly.

(4)

Isotones

Atoms which have different atomic numbers and atomic masses but the same number of neutrons are called isotones.

Example

${}^6_{14}\text{C}$, ${}^7_{15}\text{N}$ and ${}^8_{16}\text{O}$ are isotones since each contains eight neutrons.

Isobars

The atoms which have same mass numbers but different atomic numbers are called isobars.

Example

${}^{18}_{40}\text{Ar}$, ${}^{19}_{40}\text{K}$ and ${}^{20}_{40}\text{Ca}$ are isobaric atoms.

LONG ANSWERS

(1)

Nuclear Reactors

It has been possible to control fission chain reaction so that energy is released slowly at a usable rate. This controlled fission is carried out in a specifically

designed plant called a **nuclear reactor**.

Nuclear reactors are used in the production of heat, mechanical and electrical energy, radioactive isotopes, weapons materials and nuclear research.

Working Mechanism

In a nuclear reactor, controlled fission generates heat in the reactor core, which is transferred to a coolant. This heat produces high pressure steam that turns a turbine connected to a generator, producing electricity. Reactors used for this purpose are known as power reactors.

Components of Reactor

The chief components of a nuclear reactor are:

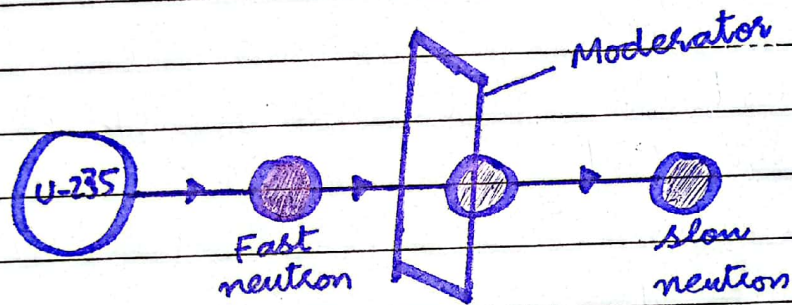
1. Fuel Elements / Core :

It consists of $U-235$ fuel rods which constitute the fuel core. The fission of $U-235$ produces heat energy and

neutrons that start the chain reaction.

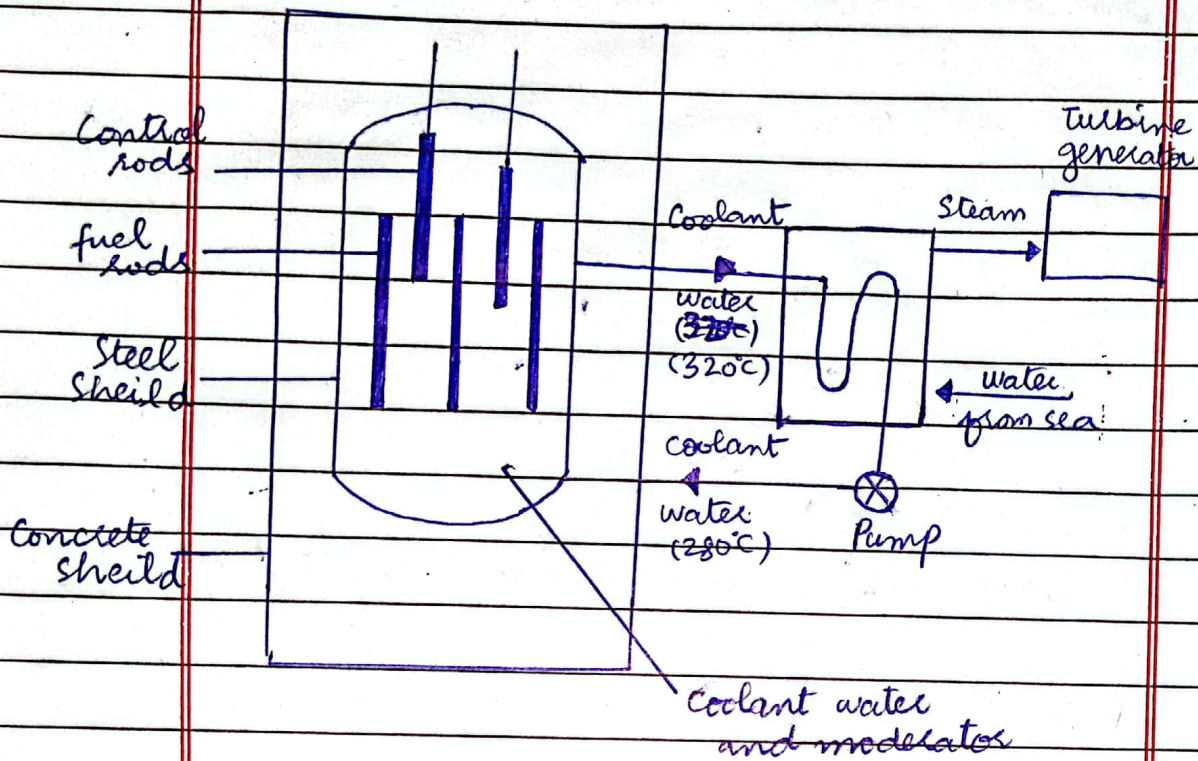
2. Moderator :

Moderator rapidly reduce the high energy of fission neutrons. The most commonly used moderator is ordinary water. Graphite rods are sometimes used. Neutrons slow down by losing energy due to collisions with atoms / molecules of the moderator.



3. Coolant :

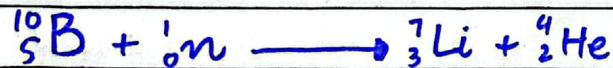
The heat generated in the core of the reactor is removed by a coolant. Both ordinary and heavy water are considered to be good coolants because of their excellent heat exchange properties. They have also the advantage of serving both as moderator and coolants.



- A light water reactor producing electricity

4. Control rods:

Control rods control the rate of fission reaction. These are made of boron-10 or cadmium, that absorbs some of the slowed neutrons.



Thus the chain reaction is prevented from going too fast -

5. Concrete Shield:

It protects the operating personnel and environment from destruction in case of leakage of radiation.

(2)

Uses of Tracers in Chemistry

Various uses of tracers are discussed as under:

1. In Quantitative Analysis:

Isotopic dilution analysis determines low-concentration substances by mixing them with a known amount of their isotopically labeled form. Changes in the labeled-to-unlabeled ratio indicates dilution by the original substance. Measuring this ratio change allows accurate quantification of the substance. This method exemplifies the use of tracers in quantitative analysis.

2. Reaction Mechanism:

Radioactive tracers may also be used in the study of mechanism of reaction. Radiocarbon, ^{14}C , is extensively used in the study of mechanism of organic reactions. There are a number of reactions in which carbon atom or groups of carbon atoms migrate from one part of the molecule to another. By labelling the migrating groups with radio carbon at suitable position, it is possible to understand the course of the reactions.

3. Radiometric Titrations:

Radioactive tracers may also be used to determine the end points of titrations where the reaction product is either insoluble or extractable.

4. Diffusion:

The process of diffusion has also been studied by tracer methods. The rate of diffusion of lead has been determined by mixing radio-lead

with ordinary lead and allowing the mixture to diffuse into inactive lead. It is found that at 473K, diffusion of lead is about ten times slower than that of tin in lead.

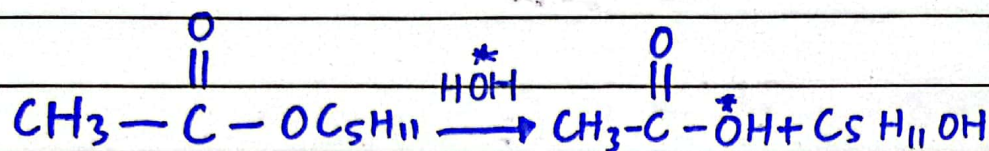
5. Research Applications:

Stable as well as radioactive isotopes are frequently used in tracing out the correct path of chemical reactions, some of which are described below:

(a) If radioactive sulphur-35 ion is added to a saturated solution of cobalt sulphide, the solid becomes radioactive. This indicates that sulphide ion exchange occurs between solid and solution in solubility equilibrium.



(b) The mechanism of ester hydrolysis can be studied by making use of labeled water (H_2^{18}O). Theoretically an ester may be hydrolyzed in two different manners to form an acid and alcohol.



The labeled oxygen appeared only in the acid, showing that $-\text{OC}_5\text{H}_{11}$ group is substituted by $-\overset{*}{\text{O}}\text{H}$ in the hydrolysis.

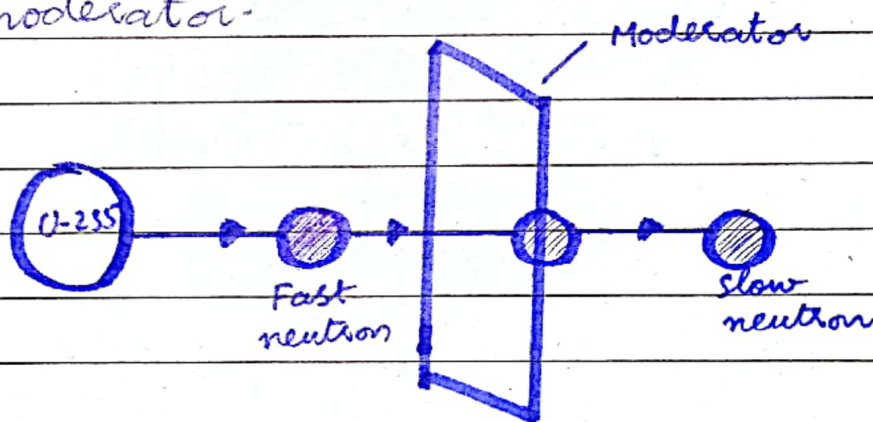
2020

Short Answers

(1)

Function of Moderator in nuclear Reactor

In a nuclear reactor, moderator rapidly reduce the high energy of fission neutrons. Neutrons slow down by losing energy due to collisions with atoms or molecules of the moderator.



The most commonly used moderator is ordinary water, although graphite is sometimes used. The most efficient moderator is helium and the next most efficient is heavy water, but this is so expensive that it has been used chiefly in research reactor.

(2)

Units of radiation exposure:

The radiation exposure is equivalent to the energy deposited in a kilogram of a substance by the radiation.

→ It is measured in an international (SI) unit called gray (Gy).

The gray is a large unit and for normal radiation protection level a series of prefixes are used

- nanogray (nGy)
- microgray (μ Gy)
- milligray (mGy)

(3)

Three applications of Tracers in chemistry:

The applications of tracers in chemistry are discussed below:

1. Radioactive tracers are used in the study of mechanism of reaction.
2. The process of diffusion has also been studied by tracer methods.

3. Isotopic tracers help determine the rate of individual steps in complex reactions.

4. In quantitative analysis, tracers serve as internal standards to precisely measure substance concentration in complex mixture.

5. Tracers help elucidate reaction pathways and identify intermediates in chemical reactions.

LONG ANSWERS

(1)

Nuclear Reactor

(Repeat 2023)

2019

Short Answers

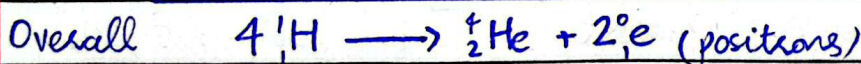
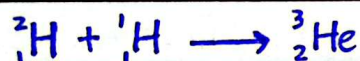
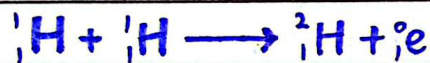
(1)

Stellar Energy:

The energy produced by stars through nuclear fusion is called stellar energy.

- In the core of stars, hydrogen nuclei fuse to form helium, releasing energy.
- This energy radiates outward, supporting the stars structure and emitting light and heat.

- In smaller stars like sun, the primary fusion process is proton-proton chain which is discussed below:



(2)

Radioactive Equilibrium:

When the rate of formation of product is equal to the rate of disintegration of

of the product then the equilibrium obtained is called the radioactive equilibrium.

This stage may be mathematically designated as

$$\frac{dN_1}{dt} = \frac{dN_2}{dt} = \frac{dN_3}{dt}$$

For Example :

In the uranium-238 decay series, uranium-238 decays to thorium-234. Radioactive equilibrium is reached when the rate at which uranium-238 forms thorium-234 equals the rate at which thorium-234 decays into protactinium-234.

(3)

Decay Constant :

Decay constant of a radioisotope describes the number of nuclide that decays per unit time.

- A larger decay constant suggests a radioisotope decays faster.

- The decay constant is given by

$$\lambda = \frac{\ln 2}{t_{1/2}}$$

- The decay constant (λ) is inversely proportional to a nuclide's half life.

A larger decay constant means a nuclide takes shorter time to reach 50% of its initial quantity, & therefore has a shorter half-life.

(4)

Equation to calculate half life of a radioactive substance:

The half life of the radioactive substance can be calculated by using the following relation:

$$t_{1/2} = \frac{0.693}{\lambda}$$

The value of λ can be found experimentally by finding the number of disintegrations per second.

Long Answers

(1)

Atomic Bomb

Atomic Bombs are nuclear weapons that use the energetic output of nuclear fission to produce massive explosions.

History:

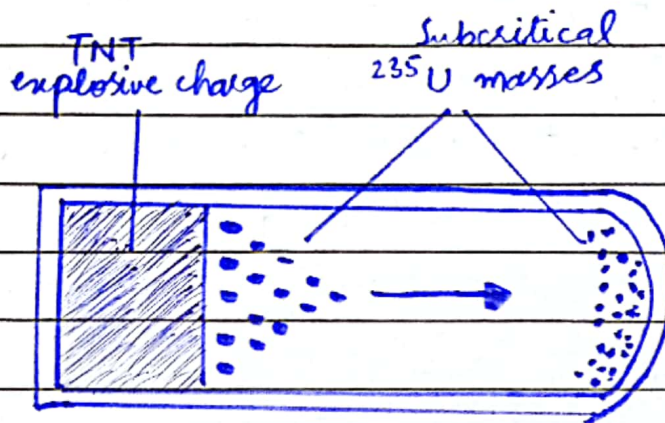
Only two nuclear weapons have been used in the course of warfare, both by the United States near the end of World War II. On August 6th, 1945, a uranium gun-type fission bomb code-named "Little Boy" was detonated over the Japanese city of Hiroshima. Three days later, on August 9th, a plutonium implosion-type fission bomb code-named "Fat Man" was exploded over Nagasaki, Japan. These bombs extensively damaged the two towns and killed approximately 200,000 Japanese people - mostly civilians. In an actual practice temperature from this bomb arises upto $10 \text{ million}^{\circ}\text{C}$ & pressure upto several million atmosphere.

Principle:

It works on the principle of fast neutron chains. If the mass of uranium is much greater than the critical mass, then the chain reactions proceed at a rapid speed and a huge explosion is produced.

Working:

A design of atomic bomb is shown below:



- A design in atomic bombs to bring together two subcritical masses of ^{235}U .

It contains two subcritical masses of fissionable material, ^{235}U or ^{239}Pu . It has a mass of trinitrotoulence (TNT) in a separate pocket. When TNT is detonated, it drives one mass of ^{235}U into the other. A super critical mass of the fissionable material is obtained. As a result of the instantaneous chain reaction, the bomb explodes with the release of tremendous amount of energy.

- Temperature developed in atomic bomb is believed to be 10 million $^{\circ}\text{C}$
- Besides many radionuclei and heat, deadly gamma rays are released.

These play havoc with life and environment - If the bomb explodes near the ground, it raises tons of dust into the air. The radioactive material adhering to dust known as **fall out**. It spreads over wide areas and is a lingering source of radioactive hazards for long periods.

(2)

Radiation Hazards

"Radiation hazards refer to the risks and dangers associated with exposure to ionizing radiation, which can cause severe health effects and environmental damage."

Sources of Radiations

Radiations can originate from natural and artificial sources:

- **Natural Sources:**

These include cosmic rays, radioactive elements in

the Earth's crust (such as uranium and radon), and naturally occurring radioactive isotopes in the human body.

• Artificial Sources:

Human activities produce ionizing radiations such as medical procedures (X-rays, CT scans), nuclear power plants, industrial processes and nuclear weapons.

Effects: & ~~R~~

Exposure to ionizing radiation can have adverse health effects:

• Acute Effects:

High doses of radiations delivered over a short period can cause acute radiation syndrome, characterized by symptoms like nausea, vomiting, hair loss, and skin burns. In severe cases, it can lead to organ failure & death.

• Chronic Effects:

Prolonged exposure to lower dose of radiation can increase the risk of developing cancer, such as thyroid cancer, leukemia etc. Genetic mutations and reproductive effects are also potential long-term consequences.

• Environmental Impact:

Radioactive contamination can ~~affect~~ affect air, water and soil, posing risks to human health and ecosystems.

Protective Measures

Several measures can be implemented to mitigate radiation hazards:

- **Time:** Reducing "the amount of time spent near radiation sources decreases exposure."

- **Distance:**

Increasing the distance from a radiation source significantly decreases exposure.

- **Shielding:**

Using appropriate shielding materials, such as lead or concrete, can absorb radiation and protect individuals.

2022

Short Answers

(i)

Why mass of an atom is less than the sum of masses of its component particles?

The mass of an atom is less than the sum of masses of its component particles due to **mass defect**. This occurs because some of the mass is converted into **binding energy** when the nucleus is formed, according to Einstein's equation $E = mc^2$. This binding energy holds nucleus together, and its equivalent mass is missing from the total atomic mass.

Example:

A Helium-4 atom has a calculated mass of 4.032 amu from its particles, but its actual mass is only 4.0026 amu. The difference (0.0294 amu) is the mass converted into binding energy.

(ii)

Natural Radioactivity

- Emission of radiation due to self disintegration of a nucleus.
- Alpha, beta and gamma radiations are emitted.
- It is a spontaneous process.
- Exhibited by elements with atomic number more than 83.
- This can't be controlled.

Artificial Radioactivity

- Emission of radiation due to disintegration of nucleus through induced process.
- Mostly elementary particles such as neutron, positron etc are emitted.
- It is an induced process.
- Exhibited by elements with atomic number less than 83.
- This can be controlled.

(iii)

Examples of uses of radioactive tracers in analytical chemistry

- Radioactive isotopes like Carbon-14 are

used in isotope dilution analysis to accurately measure the concentration of active ingredients in pharmaceutical formulations.

- Iodine-131 labeled iodide detects the end points of titration.
- Phosphorus-32 is used to detect and quantify trace amounts of phosphorus in samples.
- Iodine-131 traces contaminants to optimize water treatment processes.

(iv)

Harmful effects of radioactivity:

- Radiations can disrupt the structure and function of cells, leading to cell death and mutation. These mutations can lead to cancer or other health problems.
- High doses of radiation in a short period can cause acute radiation syndrome, with symptoms like nausea, vomiting, hair loss etc. In severe cases, it can be fatal.
- Long-term exposure to radiation can increase the risk of developing cancer.

Ayesha
Rasool
ul
Ghade

(5)

Mechanism of stellar fusion reaction of hydrogen

Repeat 2019

LONG ANSWERS

(1)

Distinguish fusion & fission with examples

Nuclear Fission	Nuclear Fusion
Definition	
Nuclear fission is the splitting of a heavy, unstable nucleus into two or more smaller, lighter nuclei	Nuclear fusion is the process of combining two light nuclei to form a heavier nucleus, releasing large amount of energy.
Energy Release	
Releases significant amount of energy, but less than fusion.	Release much more energy per reaction compared to fission.

Nuclear Fission

Nuclear Fusion

Temperature

It does not require high temperature.

Extremely high temperature is required for fusion to take place.

Control

It can be controlled and energy released can be used for peaceful purposes.

It can not be controlled and energy released can not be used properly.

By Product

Produces radioactive waste that requires long term management and storage.

Produces minimal radioactive waste which is non-toxic.

Fuel Source

Common fuels include uranium-235 and plutonium-239.

Common fuels include isotopes of hydrogen, such as deuterium & tritium.

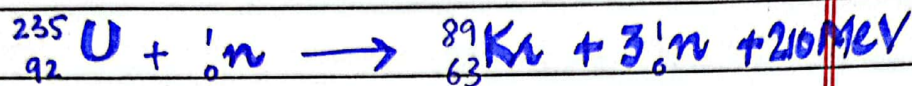
Examples

1. Nuclear Fission:

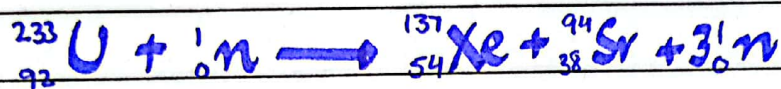
The examples

of nuclear fission are given below:

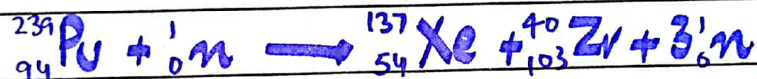
1. Splitting of Uranium - 235.



2. Splitting of Uranium - 233.



3. Splitting of Plutonium - 239

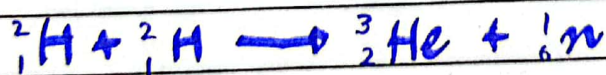


Nuclear Fusion:

1. Deuterium - Tritium Fusion



2. Deuterium - Deuterium fusion



3. Fusion of Li with a proton

